

Chapter 20 — Complications of Urologic Surgery — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Clavien-Dindo grades treatment, not severity**; transfusion = II; dialysis = IVa.
- **Compartment syndrome**: fasciotomy >30 mmHg symptomatic; >50 mmHg >3–4 h irreversible.
- **Long lithotomy case** → lower legs every 2 h.
- **Delayed ureteric injury**: repair ≤7 days or after 6 weeks; stent 4–8 weeks.
- **Umbilical Veress** → left common iliac vein commonest vessel injured.
- **Gas embolism**: desufflate → 100% O₂ → Trendelenburg + left lateral → central line.
- **Post-partial nephrectomy pseudoaneurysm** 2–3 weeks → selective angioembolisation.
- **Post-laparoscopy bowel injury**: leucopenia, single port-site pain, 24–48 h.
- **Bipolar** = energy of choice; monopolar spread >20 mm.
- **Caprini**: 0 very low, 1–2 low, 3–4 moderate, ≥5 high.

INTRODUCTION

- **Laparoscopic risk**: operating time >4 h; creatinine >1.5 mg/dL; complex case.
- **Complications** up to 9× with complexity; 4× lower with experience.
- **Robotic prostatectomy learning curve** ≈40 vs open 100–1000.

CLASSIFICATION OF COMPLICATIONS

- **Gold standard**: Clavien-Dindo, graded by treatment needed.
- **I**: bedside drugs/physio; wound opened at bedside.
- **II**: other drugs, transfusion, TPN (total parenteral nutrition).
- **IIIa** no GA vs **IIIb** under GA (general anaesthesia).
- **IVa** single organ (incl. dialysis); **IVb** multiorgan; TIA excluded.
- **V** death; suffix “d” = disability at discharge.
- **Limits**: no intraoperative events, no early/late split, weak in paediatrics.

PHYSIOLOGIC COMPLICATIONS

- **Pneumoperitoneum** → bradycardia, hypercarbia, acidosis, ↑ airway pressure.

POSITIONAL AND NEUROMUSCULAR

- **Incidence** <3% (robotic ≈6%); permanent deficit very rare.
- **Lithotomy**: peroneal (fibular neck) → foot drop; posterior tibial → weak plantarflexion.
- **Exaggerated lithotomy** → obturator → weak adduction.
- **Arm abduction >90°** → brachial plexus; axillary roll in flank/prone.
- **Arm off board**: supination → radial (wrist drop); pronation → median.
- **Ulnar**: tucked arms, elbow 90° → claw hand (4th–5th).
- **Prone**: anterior tibial → foot drop; lateral femoral cutaneous → thigh numbness.
- **Psoas hitch**: genitofemoral entrapment; sutures longitudinal along tendon.
- **Obturator transection** → 7-0 non-absorbable tension-free repair.
- **Flank incision** from 11th rib upper border/12th tip; layered closure.
- **Lower-limb compartment syndrome**: 1 in 3500; 1 in 500 cystectomies.
- **Exclude DVT** (deep vein thrombosis) by duplex.
- **Gluteal compartment syndrome**: ↓ vibration sense, pain on stretch → early fasciotomy.

URINE LEAKS AFTER UPPER TRACT SURGERY

- **Commonest partial nephrectomy complication after haemorrhage**; robotic <1%.
- **Risk**: size, endophytic, hilar, warm ischaemia, operating time.

- **Presents** ≈day 13 (3–32).
- **Gold standard diagnosis**: CT urography with delayed images.
- **Drain creatinine >48 h** confirms leak, not site.
- **Order**: Foley → stent → drain off suction/withdraw → IR drain → nephrectomy last.
- **Prolonged** = >30 days; may heal at ≥80 days.

URETERAL INJURIES

- **Hysterectomy** <1%; most unrecognised; late recognition ≈10× nephrostomy need.
- **Prophylactic ureteric catheters**: no robust evidence.
- **Landmark**: pelvic brim over common iliac artery.
- **Partial tear** → 4-0/5-0 absorbable over stent.
- **Upper two-thirds** → spatulated ureteroureterostomy.
- **Lower** → reimplant + psoas hitch; extensive loss → Boari flap.
- **Sepsis uncontrolled** → immediate repair, any timing.
- **Delayed diagnosis**: CT with IV contrast, delayed whole-ureter images.
- **Stricture >2 cm** or failed endoscopy → surgery.
- **Long upper stricture** → ileal ureter; right side → appendix.
- **Double-dye**: orange = ureterovaginal; blue = vesicovaginal.

LOWER TRACT URINARY COMPLICATIONS

- **Bladder repair**: two layers absorbable; catheter 7–10 days.
- **Diagnosis**: CT cystography.
- **Intraperitoneal** → repair; small extraperitoneal → catheter only.
- **Commonest urological**: posterior bladder-neck buttonhole at RARP (robotic radical prostatectomy).
- **Bladder cuff leak** → drain 7–10 days.
- **Anastomotic leak**: days 7–10; up to ≈15%; usually posterior.
- **Haematocrit fall >6 points** predicts leak.
- **Barbed vs monofilament**: no difference.
- **Catheter**: side-fenestrated; low intermittent suction; never continuous.
- **Formal re-repair** only early; rarely after day 7.

VASCULAR COMPLICATIONS

- **Commonest intraoperative complication**; commonest cause of conversion.
- **MIS vascular injury** 2–3%; access injury ≈0.1%.
- **Postoperative bleeding** mostly unrecognised small vessels.
- **Most reliable rescue**: suture repair; clips judiciously.
- **Benchmark**: react ≈22 s, control <1 min, repair ≈3 min.
- **Blood up Veress** → stop insufflating; leave needle/trocar in.
- **Embolism signs**: ↓ end-tidal CO₂, mill-wheel murmur, hypoxia.
- **Insufflation**: 15–18 mmHg for trocars → 12–15 mmHg.
- **AirSeal**: all trocars in before switching mode.
- **Bleeding**: 20–25 mmHg ≤10 min; venous repair at 20 mmHg.

- **Veress** 45° normal BMI (body mass index), 90° obese.
- **Skin-retroperitoneum** ≈6 cm; thin ≈2 cm.
- **AAA** (abdominal aortic aneurysm) → left upper quadrant entry.
- **FDA**: 81% trocar deaths = major vessel injury.
- **Hasson** safest for novices/adhesions; slowest.
- **Epigastric** 0.2–2%; lateral ports ≥5.5–6 cm from midline.
- **Epigastric repair**: fascial closure device figure-of-eight.
- **Ligatable**: internal iliac, inferior mesenteric only.
- **SMA** (superior mesenteric artery) suspected if medial to aorta/ anterior to renal vein.
- **Proximal SMA injury**: mortality up to 75%.
- **Stable non-expanding haematoma** → observe; expanding → open.
- **External iliac vein tear**: ↑ pressure + switch off SCDs (sequential compression devices).
- **Right renal vein** ≤2 cm → clamp; left up to 10 cm.
- **Bleeding despite clamping** → unclamp vein.
- **Stapler malfunction** ≈1%; commonest cause tissue too thick.
- **Hem-o-lok failure** = dislodgement.
- **Pseudoaneurysm/AVF** 2–3%, 2–3 weeks; <2–4 mm resolve.
- **High suspicion** → angiography direct, bypass CT.
- **Treatment of choice**: superselective coil embolisation.

BOWEL INJURIES

- **Incidence** ≈0.2%; up to two-thirds missed intraoperatively.
- **Free gas** normal ≤7 days after laparoscopy.
- **CT** oral + IV (intravenous) contrast; rectal if rectal injury.
- **Veress** → mostly small bowel; advances 1–2 cm freely.
- **Adhesiolysis**: cold scissors, no cautery.
- **Thermal burn** → Lembert sutures; never conservative.
- **Spread**: harmonic < bipolar < LigaSure < hot shears.
- **Duodenum (right)**: <5 mm injury perforates.
- **Pancreatic tail (left)**: obvious injury → distal pancreatectomy; drain amylase.
- **Rectal injury**: laparoscopic ≈1%; open ≈ robotic; salvage ≈7%.
- **Commonest RARP rectal site**: apex, posterior dissection.
- **Repair**: two layers; bubble test; drain; antibiotics 3–7 days.
- **Salvage/irradiated injury** → diverting colostomy.
- **Unrecognised rectal injury** → exploration + diverting colostomy.

- **Port hernia** 0.3–5.4%; 12 mm ≫ 10 mm; close fascia.
- **VTE** (venous thromboembolism) highest after cystectomy (≈5%); most after discharge.
- **Preventer RCT**: heparin after prostatectomy → no VTE benefit.
- **AUA 2009 high risk**: >60 y, prior VTE, cancer, thrombophilia → drug prophylaxis.
- **Caprini**: 1 = 41–60 y; 2 = 61–74 y; 3 = ≥75 y.
- **Caprini 2**: open or laparoscopic >45 min; malignancy.
- **Pharmacological** 2–12 h preop; fondaparinux 6–8 h postop.
- **Extended LMWH** (low-molecular-weight heparin) ≥1 month after abdominopelvic cancer surgery.
- **Aspirin 160 mg** only if heparins contraindicated.
- **Lymphocele** <5 cm asymptomatic → conservative.
- **Extended node dissection** ≈2× lymphocele rate of limited.
- **Recurrent lymphocele** → laparoscopic marsupialisation.

SECOND VICTIM SYNDROME

- **Affects** ≈half of providers; cognitive impairment lasts months.

CLASSIC TRAPS

- **Trap**: transfusion = grade I — **Actually**: grade II.
- **Trap**: high early drain output = leak — **Actually**: check drain creatinine.
- **Trap**: drain on suction helps leak — **Actually**: siphon keeps fistula open.
- **Trap**: catheter traction for anastomotic leak — **Actually**: never; disrupts anastomosis.
- **Trap**: repair delayed ureteric injury at 3 weeks — **Actually**: ≤7 days or ≥6 weeks.
- **Trap**: free gas day 4 = perforation — **Actually**: normal ≤7 days.
- **Trap**: bowel injury → leucocytosis — **Actually**: leucopenia post-laparoscopy.
- **Trap**: remove bleeding trocar — **Actually**: leave in; tamponade.
- **Trap**: CT first for suspected pseudoaneurysm — **Actually**: direct angiography.
- **Trap**: aspirate lymphocele — **Actually**: reaccumulates; drain or marsupialise.